

Koustubh Verma

Portfolio: [portfolio-psi-flax.vercel.app](#)
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EDUCATION

- Kalinga Institute of Industrial Technology** Bhubaneswar, India
Bachelor of Technology in Electronics and Computer Science Engineering July 2023 - June 2027
Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases

SKILLS SUMMARY

- Languages:** Java, Python, JavaScript (Node.js), C, C++
- Core CS Fundamentals:** Data Structures & Algorithms, System Design, Object-Oriented Programming, Operating Systems, Problem Solving
- Backend & APIs:** Spring Boot, REST APIs, Microservices Architecture, FastAPI, JDBC
- Data Engineering & Databases:** Apache Kafka, Apache Spark, Airflow, ETL Pipelines, BigQuery, SQL, PostgreSQL, Redis, Data Modeling
- Infrastructure & DevOps:** Docker, Kubernetes, GCP, AWS, CI/CD Pipelines, GitHub Actions
- AI & ML:** LLMs, RAG, LangChain, Agentic AI, Machine Learning, AI-assisted Development (Cursor, Claude)

INTERNSHIP

- Synarion IT Solutions** Jaipur, India (Remote)
Frontend Developer (Part-time) May 2025 - Jul 2025
 - Education LMS:** Developed responsive Education LMS using MERN stack with scalable frontend components and optimized performance.
 - API & UI/UX:** Integrated RESTful APIs and enhanced UI/UX workflows in collaboration with backend developers and designers.
- Freelance Web Developer** Remote
Full Stack Developer Jan 2024 - Present
 - AI/ML Integration:** Built web applications with integrated ML models for predictive analytics and data visualization.
 - Web Development:** Deployed end-to-end projects using modern frameworks, focusing on performance optimization and SEO.
 - Client Solutions:** Managed full-cycle development from requirements to cloud deployment on AWS and Azure platforms.

PROJECTS

- Order Tracking System** (*Python, SQL, REST APIs, System Design*) [Github]
Built a high performance backend that manages the full order lifecycle across 5+ order states with complete CRUD support. Normalized the database schema which brought query execution time down by 40%. Kept the codebase modular so new features could be added without breaking existing ones.
- Seva-AI — Real-Time Volunteer Deployment Platform** (*TypeScript, Node.js, Real-time Systems*) [Github]
Engineered a resilient backend that handles concurrent data streams from 100+ field inputs with low latency. Built data integrity pipelines and designed system components around clean architecture principles to support real-time cross-functional deployment decisions.
- LegalMind AI — Multi-Document Legal Assistant** (*Flask, Microservices, Python*) [Github]
Designed a production backend with 14 loosely coupled microservices handling automated risk assessment across complex legal document portfolios. Built cross-document relationship detection that reduced manual review time by over 60%.
- ContextIQ — AI Personalized Shopping Assistant** (*Agentic RAG, Node.js, ChromaDB, Gemini 2.5 Pro*) [Github]
Built a context-aware product recommendation engine using Agentic RAG and multimodal understanding, directly applicable to e-commerce personalization at scale. Integrated persistent memory and a vector store that improved recommendation accuracy by 35% over baseline retrieval.
- AI Shiksha — Adaptive Learning Platform** (*React, NestJS, PostgreSQL, REST APIs*) [Github]
Built a full stack education platform serving personalized learning paths to multiple concurrent users. Designed backend APIs in NestJS with PostgreSQL that handled real-time progress tracking across 3+ user roles. Reduced average content load time by 30% through query optimization and response caching.

LEADERSHIP EXPERIENCE

- Video Editing Lead (KIIT Film Society):** Lead creative direction for society video content. Manage editing team to deliver high-quality event promos and documentaries with focus on storytelling and consistent branding.
- Web Developer (KIIT Electrical Society):** Head development of society web platforms. Guide developer team from wireframes to deployment, ensuring fast, user-friendly sites for high-traffic college events.